Jacobi Field

(M,g) is a Riemannian mfd. Consider $p\in M$,

$$\exp_p : T_pM \to M,\ v \mapsto \text{a point}$$

f is a parametrized surface,

$$f(t,s) = \exp_p t v(s),\ 0\le t\le 1,\ -\varepsilon\le s\le\varepsilon,\ v(0)=v,\ v'(0)=w.$$

$$(d\exp_p)_v w = \frac{df}{ds}(1,0)$$

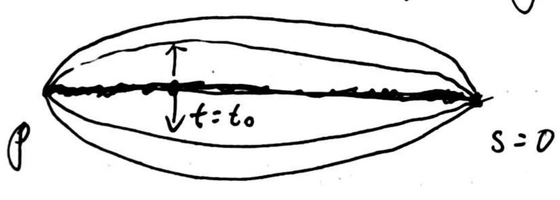
$$\frac{\partial f}{\partial s}(t,s) = (d\exp_p)_{tv(s)}\left(\frac{\partial}{\partial s}(tv(s))\right)$$
$$= d(\exp_p)_{tv(s)}(tv'(s))$$

So

$$\frac{\partial f}{\partial s}(1,0) = (d\exp_p)_v w$$

Consider $\underbrace{|(d\exp_p)_v(w)|}_{A}$, $\Rightarrow$ this is the rate of spreading of geodesic $t\mapsto \exp_p tv(s)$

t_0: $A>0$



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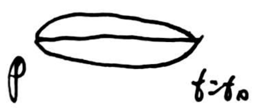
$A=0$



Then, consider

$$0=\frac{D}{\partial s}\left(\frac{D}{\partial t}\frac{\partial f}{\partial t}\right) = \frac{D}{\partial t}\frac{D}{\partial s}\frac{\partial f}{\partial t} - R\left(\frac{\partial f}{\partial s},\frac{\partial f}{\partial t}\right)\frac{\partial f}{\partial t}$$
$$= \frac{D}{\partial t}\frac{D}{\partial s}\frac{\partial f}{\partial t} + R\left(\frac{\partial f}{\partial t},\frac{\partial f}{\partial s}\right)\frac{\partial f}{\partial t}$$

With $s=0$, we have $\begin{cases} \frac{\partial f}{\partial s}(t,0) = J(t) \\ \frac{\partial f}{\partial t}(t,0) = \gamma'(t) \end{cases}$.

$$\Rightarrow \frac{D^2}{dt^2} J + R(\gamma'(t), J(t))\gamma'(t) = 0 \qquad \underline{(*)}$$

$(*)$ is call <u>Jacobi equation</u>.

Def:

Let $\gamma : [0,a] \to M$ be a geodesic in M. J is a vector field along γ. J is a Jacobi if it satisfies Jacobi equation $(*)$ for all $t \in [0,a]$.

Let $e_1(t), \cdots, e_n(t)$ be parallel, orthonormal field along γ. Then,

$$J(t) = \sum_i f_i(t) e_i(t)$$

$$a_{ij} = \langle R(\gamma'(t), e_i(t))\gamma'(t), e_j(t) \rangle, \quad i,j = 1,2,\cdots,n$$

$$\Rightarrow \frac{D^2 J}{dt^2} = \sum_i f_i''(t) e_i(t)$$

Since

(2)

$$R(\gamma', J)\gamma' = \sum_i \langle R(\gamma', J)\gamma', e_j \rangle e_j$$
$$= \sum_i f_i \langle R(\gamma', e_i)\gamma', e_j \rangle e_j$$
$$= \sum_{i,j} f_i a_{ij} e_j$$

Then, $(*)$ becomes

$$f_j''(t) + \sum_i a_{ij}(t) f_i(t) = 0, \quad j = 1,2,\cdots,n$$

which is a linear system of 2rd-order.

Given $\overset{n\text{-dim}}{J(0)}$ and $\frac{DJ}{Dt}(0)$, the equation has a unique C^∞ solution on $[0,a]$. $\exists\ 2n$ linearly independent Jacobi field along γ.

Example:

Let M be a Riemannian mfd of content sectional curvature K. $\gamma: [0,l] \to M$ be a normalized geodesic on M. Let J be Jacobi field along γ, $|\gamma'| = 1$.

Consider the Jacobi field along γ, that are normal to γ'. Then,

$$\langle R(\gamma', J)\gamma', T\rangle = K\{\langle \gamma', \gamma'\rangle\langle J, T\rangle - \underbrace{\langle \gamma', T\rangle\langle J, \gamma'\rangle}_{=0}\}$$

$$= K\langle J, T\rangle \quad \text{(where } T \text{ is any field along } \gamma)$$

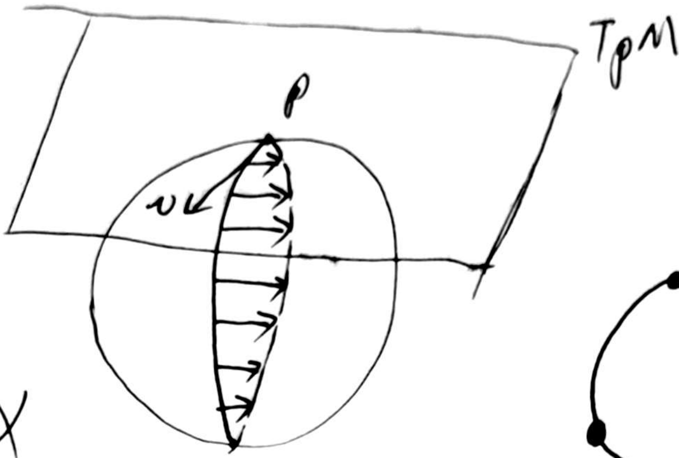
$$\Rightarrow \quad \frac{D^2 J}{dt^2} + KJ = 0$$

Let $w(t)$ be a parallel field along γ with $\langle \gamma'(t), w(t)\rangle = 0$. Let $|w(t)| = 1$.

Then, we have the 2nd-order ODE with characteristic equations $\begin{cases} \langle \gamma'(t), w(t)\rangle = 0 \\ |w(t)| = 1 \end{cases}$. So

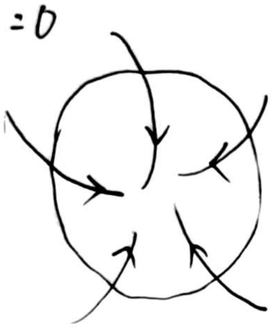
$$J(t) = \begin{cases} \frac{\sin(t\sqrt{K})}{\sqrt{K}} w(t) & K > 0 \\ t\, w(t) & K = 0 \\ \frac{\sinh(t\sqrt{-K})}{\sqrt{-K}} w(t) & K < 0 \end{cases}, \text{ initial } \begin{cases} J(0) = 0 \\ J'(0) = w(0) \end{cases}$$

① $K>0$. $K=1$. S^2

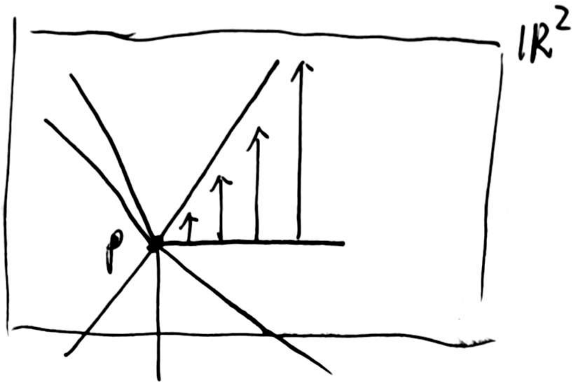


$J(t) = \sin t \; w(t)$

$J(0) = J(\pi) = 0$



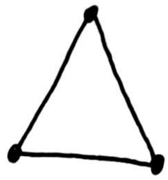
② $K=0$. $\mathbb{R}^2$



$J(t) = t\, w(t)$

$J(0)=0$, $|J'(0)|=1$

$J(t)>0$, $t>0$

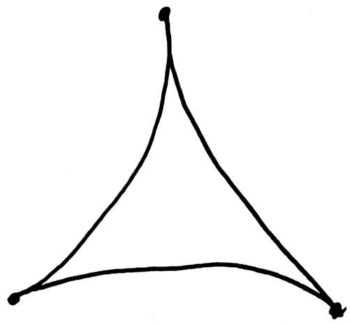


③ $K<0$, $K=-1$. $\mathbb{H}^2$

$J(t) = \sinh t \; w(t)$

$J(0)=0$

$J(t)$, $t>0$



Prop.

Let $\gamma:[0,a]\to M$ be a geodesic, J be a Jacobi field along γ with $J(0)=0$, $\frac{DJ}{dt}(0)=w$, $\gamma'(0)=v$. Consider w as an element of $T_{av}(T_{\gamma(0)}M)$.

Construct a curve $v(s)$ in $T_{\gamma(0)}M$ with $v(0)=av$, $v'(0)=w$.

Put $f(t,s)=\exp_p(\frac{t}{a}v(s))$, $p=\gamma(0)$. Define a Jacobi field $\bar{J}(t)=\frac{\partial f}{\partial s}(t,0)$.

Claim: $\bar{J}=J$ on $[0,a]$.

Proof:

For $s=0$, we have

$$\frac{D}{dt}\frac{\partial f}{\partial s}=\frac{D}{dt}((d\exp_p)_{tv}(tw))=\frac{D}{dt}(t(d\exp_p)_{tv}(w))$$

$$=(d\exp_p)_{tv}(w)+t\frac{D}{\partial t}((d\exp_p)_{tv}(w))$$

$$\overset{t=0}{\Longrightarrow}\frac{D\bar{J}}{dt}(0)=\frac{D}{dt}\frac{\partial f}{\partial s}(0,0)=(d\exp_p)_0(w)=w.$$

$$\Rightarrow\begin{cases}J(0)=\bar{J}(0)\\ \frac{DJ}{dt}(0)=\frac{D\bar{J}}{dt}(0)=w\end{cases}$$

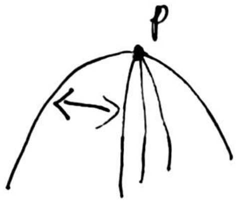
$$\Rightarrow J=\bar{J}.$$

5 $\square$

Then, we can know

$$J(t)=(d\exp_p)_{t\gamma'(0)}(tJ'(0)),\ t\in[0,a],\ v=\gamma'(0),\ w=J'(0),\ J(0)=0.$$

Next, we consider the speed of spread apart.



Prop.

$p\in M$, $\gamma:[0,a]\to M$ is a geodesic with $\gamma(0)=p$, $\gamma'(0)=v$. Let $w\in T_v(T_pM)$ with $|w|=1$ and J be Jacobi field along γ,

$$J(t)=(d\exp_p)_{tv}(tw),\ 0\le t\le a.$$

Claim: $|J(t)|^2=t^2-\frac{1}{3}\langle R(v,w)v,w\rangle t^4+o(t^4)$.

Proof:

$$\langle J, J\rangle(0) = 0 \qquad 0\cdot t^0$$

$$\langle J, J\rangle'(0) = 2\langle J, J'\rangle(0) = 0 \qquad 0\cdot t$$

$$\langle J, J\rangle''(0) = 2\langle J', J'\rangle(0) + 2\langle J'', J\rangle(0) = 2\times 1 = 2 \qquad 1\cdot t^2$$

As $J''(0) = -R(\gamma', J)\gamma'(0) = 0$,

$$\langle J, J\rangle'''(0) = 6\langle J', J''\rangle(0) + 2\langle J''', J\rangle(0) = 0 \qquad 0\cdot t^3$$

Then, we compute $\langle J, J\rangle''''(0)$. We know

$$\frac{D}{dt}(R(\gamma', J)\gamma')(0) = R(\gamma', J')\gamma'(0)$$

Take any vector field W.

$$\frac{d}{dt}\langle R(\gamma', J)\gamma', W\rangle = \langle \frac{D}{dt}(R(\gamma', J)\gamma', W\rangle + \langle R(\gamma', J)\gamma', W'\rangle$$

$$\|$$

$$\frac{d}{dt}\langle R(\gamma', W)\gamma', J\rangle = \underbrace{\langle \frac{D}{dt}(R(\gamma', W))\gamma', J\rangle}_{=0 \text{ when } t=0} + \langle R(\gamma', W)\gamma', J'\rangle - \underbrace{\langle R(\gamma', W')\gamma', J\rangle}_{=0}$$

$$\Rightarrow \langle \frac{D}{dt}(R(\gamma', J))\gamma', W\rangle = \langle R(\gamma', J')\gamma', W\rangle$$

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$$\Rightarrow J'''(0) = -R(\gamma', J')\gamma'(0)$$

Then,

$$\langle J, J\rangle''''(0) = 8\langle J', J'''\rangle(0) + 6\langle J'', J''\rangle(0) + 2\langle J'''', J\rangle(0)$$

$$= -8\langle J', R(\gamma', J')\gamma'\rangle(0)$$

$$= -8\langle R(v, w)v, w\rangle \rightarrow -\frac{1}{3}\langle R(v, w)v, w\rangle\cdot t^4$$

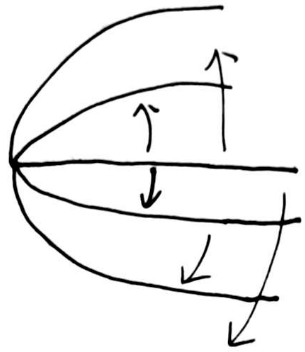
So

$$|J(t)|^2 = t^2 - \frac{1}{3}\langle R(v,w)v, w\rangle t^4 + o(t^4).$$

$\square$

Let p be the initial point of γ and $\sigma = \operatorname{span}\{v, w\} \subset T_pM$. For the sectional curvature K, we have

$$|J(t)|^2 = t^2 - \frac{1}{3}K(p,\sigma)t^4 + o(t^4)$$
$$= t^2(1 - \frac{1}{3}K(p,\sigma)t^2 + O(t^4))$$



$$\Rightarrow |J(t)| = t(1 - \frac{1}{6}K(p,\sigma)t^2 + o(t^2)) \quad (\sqrt{1+\varepsilon} = 1 + \frac{1}{2}\varepsilon - \frac{1}{8}\varepsilon^2 + \cdots)$$
$$= t - \frac{1}{6}K(p,\sigma)t^3 + o(t^3)$$

Example:

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$\mathbb{R}^n$ ① $\begin{cases} |J(0)| = 1 \\ |J'(0)| = 0 \end{cases}$ ② $\begin{cases} |J(0)| = 0 \\ |J'(0)| = 1 \end{cases}$

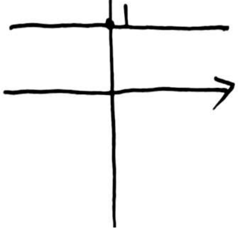
$$\frac{D^2 J}{dt^2} = 0$$

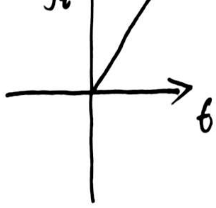
$e_1(t), \cdots, e_n(t)$

$$J(t) = \sum_i f_i(t) e_i(t)$$

$$f_i''(t) e_i(t) = 0$$

① $\begin{cases} f_i(0) = 1 \\ f_i'(0) = 0 \end{cases}$ ② $\begin{cases} f_i(0) = 0 \\ f_i'(0) = 1 \end{cases}$





$f_i(t) = t$

Def (Conjugate point)

$\gamma:[0,a]\to M$ be a geodesic. The point $\gamma(t_0)$ is said to conjugate $\gamma(0)$ along γ, $t_0\in[0,a]$, if

$\exists$ a Jacobi field along γ, J, s.t.

$$J(0)=J(t_0)=0$$

The maximal number of linearly independent field is called <u>multiplicity</u> of the conjugate point $\gamma(t_0)$.

For $\dim M=n$, $\exists$ n linearly independent Jacobi fields along the $\gamma:[0,a]\to M$, $J(0)=0$. The Jacobi fields $J_1,\cdots,J_k$, $J_i(0)=0$ are linearly independent iff $J_1'(0),\cdots,J_k'(0)$ are linearly independent.

$$J(t)=t\gamma'(s)\neq 0,\ \text{for } t\neq 0.\ \boxed{n-1}$$

$$\text{multiplicity}\leq n-1$$

Example: 8

S^n, $K=1$. $J(t)=\sin t\, w(t)$. $J(0)=J(\pi)=0$.

multiplicity $=n-1$

$C(p)=-p$.

Def:

The set of conjugate point to the point $p \in M$, for all geodesic that start at p, is called the conjugate Locus of p and is denoted by $C(p)$.

Prop:

$\gamma : [0,a] \to M$ is a geodesic and $\gamma(0) = p$. $q = \gamma(t_0)$, $t_0 \in [0,a]$ is conjugate to p along γ iff $v_0 = t_0 \gamma'(0)$ is a critical point of $\exp_p$.

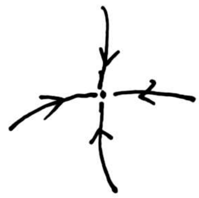
critical point:

$\exp_p : T_pM \to M$

$(d\exp_p)_v : T_v(T_pM) \to T_{\exp_p(v)}M$ (not surjective)

$\exists\, w \in T_pM, w \neq 0$, s.t.

$d(\exp_p)_v(w) = 0$



Multiplicity of q is equal to the dim of kernal of the linear map $(d\exp_p)_{v_0}$.

$$\text{multi } q = \dim \ker(d\exp_p)_{t_0 v}$$